

SUJAY KUMAR REDDY KATREDDY

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Summary

Backend developer with 4+ years of experience building APIs, microservices, and database-driven applications, improving system efficiency by 30%, and sustaining 99.9% uptime. Passionate about scalable server-side solutions, I apply structured problem-solving and Agile methods to deliver reliable, secure, and high-performance systems. My career goal is to advance in backend engineering, and employers should hire me for my proven ability to deliver measurable results that enhance business operations.

Education and Training

Master of Science: Data Science

University At Buffalo, NY

- Graduate Teaching Assistant: Distributed Systems, Data Structures & Algorithms

Skills

- API Development
 - RESTful Services
 - Graph's
 - Microservices Architecture
 - Database Management
 - SQL Optimization
 - NoSQL Databases
 - Data Modeling
 - Server-Side Programming
 - Authentication and Authorization
- Caching Strategies
 - Message Queues
 - Containerization
 - Orchestration
 - CI/CD Pipelines
 - Cloud Deployment
 - Problem-Solving
 - Debugging and Error Handling
 - Team Collaboration
 - Technical Documentation

Experience

Software Development Engineer

Toast Inc

08/2024 to Current

Boston MA

- Built backend systems for Sous Chef, Toast's AI-driven insight engine, enabling over 10,000 restaurant operators to query sales, labour, and food-cost data; increased reporting efficiency by 25%.
- Converted a monolithic architecture into Kubernetes microservices with defined API contracts, reducing downtime by 30% and improving scalability across distributed systems.
- Engineered a Redis-based asynchronous prefects layer to process repeat queries, lowering query latency by 40% during peak traffic periods.
- Developed a fallback orchestration pipeline with cached and pre-ranked insight cards, sustaining 99.98% system uptime during large language model degradation.
- Implemented role-based API layers, PII masking, audit logging, and Kafka-based delivery pipelines with LaunchDarkly toggles, ensuring SOC 2 and PCI-DSS compliance while lowering rollout risks by 20%.

Software Development Engineer

Flipkart

06/2021 to 07/2023

Bengaluru India

- Migrated the Order Management System from monolithic to event-driven microservices with Kafka and Protobuf, handling 5K+ QPS and maintaining 99.99% uptime during Big Billion Days.
- Developed idempotent order placement logic with dedupe tokens and hash keys, preventing duplicate orders and lowering failure rates by 18%.
- Implemented retry logic with exponential backoff, circuit breakers, and DLQs, increasing retry success rates by 25% in peak traffic scenarios.
- Integrated OpenTelemetry tracing for cross-service observability, reducing debugging time by 50% and improving incident resolution.
- Designed warehouse routing logic with dynamic allocation, cutting Tier-2 fulfilment failures by 20% and decreasing logistics costs by 12%.
- Built a visual search engine with CLIP embeddings and Faiss ANN indices, achieving sub-150 ms query response across 15M+ SKUs and supporting 8.5M daily queries.
- Deployed multimodal fallback using BERT and metadata fusion, raising cold-start coverage from 76% to 91.3% for new catalogue items.
- Developed real-time indexing pipelines with Kafka, ensuring <5s data freshness and improving product search reliability by 30%.
- Implemented personalisation re-rankers using clickstream data, boosting click-through rate by 22.7% and increasing average order value by 5.8%.
- Coordinated multi-zone canary deployments, sustaining 99.99% availability while scaling services for peak festival traffic.

Software Development Engineer Intern

Flipkart

01/2021 to 06/2021

Bengaluru India

- Developed the seller onboarding portal as a React single-page application with validation logic and guided workflows, improving form accuracy by 25%.
- Optimised onboarding processes for Tier 2/3 sellers, increasing activation rates by 36% in two quarters and lowering abandonment rates.
- Integrated Aadhaar, PAN, GST, and bank verification APIs with retry and failover mechanisms, sustaining 99.5% system uptime.
- Automated compliance verification scripts, reduced manual review tickets by 18% and shortened processing times.

Project

PregniCare AI — AI-powered health support platform

- Built a Flask-based AI assistant with Gemini API to deliver trimester-specific guidance, secure onboarding, and newborn care tips, reduced external info lookup by 60% via chatbot-driven contextual support

E-commerce API — scalable order management and admin dashboard

- Designed a role-based order management system using Flask, MongoDB, Redis, and Celery with secure Stripe/PayPal integrations and React admin UI; handled 500K+ concurrent users while automating refunds, tracking, CSV uploads, and inventory sync with async flows